

Medlyn Sairah Jacob

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EDUCATION

Binghamton University, State University of New York, Thomas J. Watson College of Engineering and Applied Science

MS in Computer Science, Artificial Intelligence Track | GPA: 3.587/4.00

Expected December 2026

Relevant Coursework: Machine Learning, Artificial Intelligence, Computer Architecture, Data Structures & Algorithms

Mar Baselios College of Engineering and Technology, Trivandrum, India

BTech in Computer Science and Engineering | GPA: 3.60/4.00 | Honors in Machine Learning

May 2024

TECHNICAL SKILLS

Programming Languages: Python, Java, C, JavaScript, SQL, PowerShell

Artificial Intelligence & Machine Learning: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Prompt Engineering, Model Evaluation, Feature Engineering, Scikit-learn, TensorFlow, PyTorch, XGBoost, YOLOv8, EfficientDet

Data Science & Analytics: Pandas, NumPy, Matplotlib, Data Analysis, Statistical Analysis, Data Visualization

Software Engineering: Object-Oriented Programming (OOP), Data Structures & Algorithms, REST APIs, API Integration, Design Patterns, Debugging, Testing, Software Development Lifecycle (SDLC)

Cloud, DevOps & Deployment: AWS SageMaker, Docker, Linux, Git, Model Deployment, ML Pipelines

Databases & Data Management: MySQL, SQL, ChromaDB, Data Modeling, Data Integration

Systems & Computer Architecture: Computer Architecture, Pipeline Simulation, Register Renaming, Branch Prediction, Performance Analysis, Operating Systems Fundamentals

Automation & Enterprise Platforms: Workflow Automation, Conversational AI, IT Service Automation, Enterprise Integrations, PowerShell Scripting

Visualization & Monitoring: Grafana, Streamlit, Plotly, Tableau, Power BI

Web Technologies: HTML, CSS, React.js, Netlify, GoDaddy Hosting

PROFESSIONAL EXPERIENCE

Signify Innovations Lab (Philips), Software Engineer

February 2024 – August 2025

- Automated enterprise IT workflows including password resets, account unlocks, and access provisioning using Python and PowerShell, reducing manual operational effort by 80%
- Designed and optimized conversational AI workflows, intent routing, validation logic, and API-based integrations on the Amelia platform, improving automated resolution rates for common IT requests
- Built observability dashboards in Grafana to monitor workflow health, operational metrics, system performance, and automation reliability across production pipelines
- Analyzed production chatbot behavior and workflow execution data to improve system reliability, response accuracy, and end-to-end automation performance

Nissan Digital LLP, Software Engineer Intern

August 2023 – February 2024

- Developed an aspect-based sentiment analysis pipeline on AWS SageMaker using LLaMA2, processing 50,000+ customer feedback entries with prompt engineering, fine-tuning, and model evaluation workflows
- Conducted iterative benchmarking of prompt strategies, inference outputs, and evaluation logic to improve multi-aspect sentiment classification performance and model generalization
- Built Streamlit dashboards and Plotly visualizations for large-scale sentiment analysis and feature-level performance monitoring across 50,000+ records

PROJECTS

MemoraOS - AI Memory Operating System | Developer

May 2026 – Present

- Built a local-first AI memory system using vector embeddings, semantic retrieval, and RAG pipelines to enable persistent contextual reasoning across codebases, documentation, and developer workflows

APEX CPU Pipeline Simulator | Developer

November 2025 – January 2026

- Designed and benchmarked a 7-stage out-of-order CPU pipeline simulator to evaluate instruction scheduling, execution latency, and system performance under simulated workloads
- Implemented register renaming, reservation stations, reorder buffer (ROB), and branch prediction mechanisms to model low-level execution behavior and system efficiency
- Built validation scenarios and test cases to analyze pipeline hazards, execution correctness, and performance bottlenecks in a simulated systems environment

UNI-Q International Logistics | Developer

November 2025 – January 2026

- Developed a React.js logistics website for a freight forwarding startup, streamlining customer interactions and improving UX
- Utilised Figma, GoDaddy Hosting, Netlify and GitHub for design, deployment and version control

SleepSense - Machine Learning Model for Sleep Apnea Detection | Developer

March 2025 – July 2025

- Built and evaluated machine learning pipelines using Scikit-learn and XGBoost to classify sleep apnea from physiological and lifestyle features

- Compared Logistic Regression, Random Forest, and XGBoost models using standard evaluation metrics, selecting the best-performing model for deployment

SightMate - Assistive Headgear for the Visually Impaired | Co-Developer*October 2022 – May 2024*

- Developed a research-based assistive AI prototype for real-time computer vision, using camera-based visual perception with YOLOv8 and EfficientDet for object and currency recognition on Raspberry Pi.
- Evaluated model performance for real-time inference and recognition accuracy, integrating voice-guided feedback to improve environment awareness for visually impaired users.
- Designed the headgear prototype in Blender and contributed to a patented assistive technology system recognized with a Government of India Design Patent.

LEADERSHIP AND INVOLVEMENT

School of Business, SUNY Binghamton | Graduate Student Assistant*March 2026 – Present*

- Assist students with Python-based business analytics coursework, including debugging, office hours support, and guidance on libraries such as NumPy, Pandas, and Matplotlib.

Watson Career and Alumni Connections, SUNY Binghamton | Student Assistant*March 2026 – Present*

- Supporting career advising operations and student-recruiter engagement at the university career center

Mar Baselios College of Engineering and Technology | Placement Coordinator*February 2023 – May 2024*

- Coordinated campus placements and student-recruiter communication for smooth recruitment logistics